

Answer for Midterm Exam: Economics 142

1. (28 pts. 7pts. each) Choose the correct letter for each question.

- (a) For any bridge (i, j) that has some neighbors,
- A. its neighborhood overlap must be 0.
 - B. its neighborhood overlap is at most 0.5.
 - C. the clustering coefficient must be 0 at both i and j .
 - D. the clustering coefficient must be at most 0.5 at both i and j .

Answer. A

- (b) The degree distribution at each node of random graph $G(n, 5/n)$ is approximately (when n is large)
- A. Poisson distribution with parameter $\lambda = 2.5$ ($P(d) = \frac{2.5^d e^{-2.5}}{d!}$ for $d = 0, 1, \dots$).
 - B. Poisson distribution with parameter $\lambda = 5$.
 - C. Power-law distribution $P(d) = cd^{-2.5}$ for $d = 0, 1, 2, \dots$
 - D. Power-law distribution $P(d) = cd^{-5}$ for $d = 0, 1, 2, \dots$

Answer. B

- (c) Suppose that there is a cluster of density 0.6 in some undirected network. Consider the threshold model where every node initially chooses B but prefers to choose A over B when and only when at least 50% of its neighbors choose A .
- A. No group of “initial adopters” of A can induce everyone in the cluster to choose A .
 - B. No group of “initial adopters” of A outside of the cluster can induce everyone in the cluster to choose A .
 - C. Any group of “initial adopters” of A can induce everyone in the cluster to choose A .
 - D. Any group of “initial adopters” of A outside of the cluster can induce everyone in the cluster to choose A .

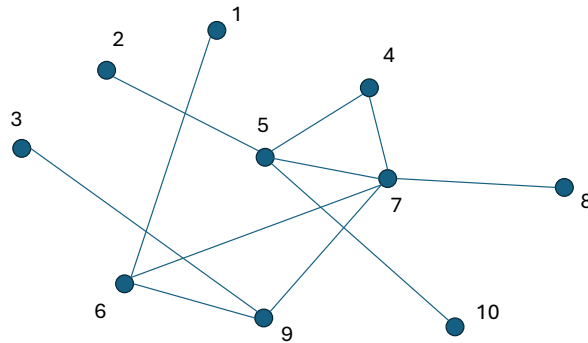
Answer. B

- (d) Consider a Degroot model with four nodes in a line network (1 – 2 – 3 – 4), where each node assigns weight 0.5 to itself and distribute the remaining weight 0.5 equally to its neighbors.
- A. The two nodes 1 and 4 at the edge have more influence over the long-run consensus than the two nodes 2 and 3 in the middle in the long run.
 - B. The two nodes 2 and 3 in the middle have more influence over the long-run consensus than the two nodes 1 and 4 at the edge.
 - C. Every node has an equal influence over the long run consensus (i.e. $w_i^* = 0.25$ for $i = 1, 2, 3, 4$).
 - D. Every node influences the long-run consensus differently.

Answer: B

(Note: Let $w_e^* = w_1^* = w_4^*$ be the long-run influence weight for node 1 and 4 and $w_m^* = w_2^* = w_3^*$ be the weight for node 2 and 3. By $w^* = w^*T$, they satisfy $1/2w_e^* + 1/3w_m^* = w_e^*$. This immediately implies $w_m^* > w_e^*$. Since the weights sum up to 1, we can actually derive them explicitly using $w_e^* + w_m^* = 1/2$ as follows: $w_e^* = 1/5$, $w_m^* = 3/10$.)

2. (36 pts.) Answer the questions about the following network.



- What is the diameter of this network?
- Find all pair of nodes (i, j) such that their distance $d(i, j)$ is equal to the diameter in (a)
- Find all the nodes where the clustering coefficient is $1/3$.
- Interpret this network as a Degroot model, where each node is influenced by all its neighbors and itself with equal weights. For example, if a node has 3 neighbors, then it assigns $1/4$ weight to the opinion of each neighbor and its own opinion. Consider the dynamics of opinions according to the updating rule of Degroot model. Do all 10 nodes reach a consensus given any initial opinions $\mathbf{x}(0) = (x_1(0), \dots, x_{10}(0))$? Explain why.

Answer.

(a) (6 pts.) 4

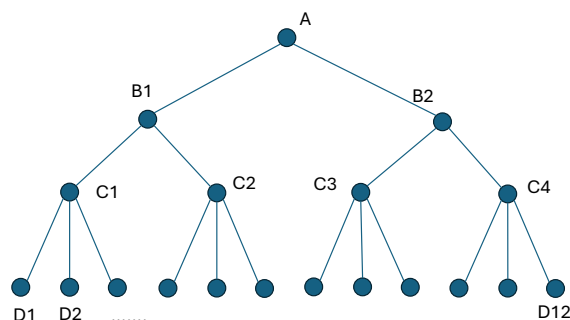
(b) (8 pts.) $(1, 2), (1, 10), (2, 3), (3, 10),$

(c) (8 pts.) 6, 9

(d) (14 pts. 4 pts. for Y/N, 5 pts. each for two sufficient conditions) Yes. They reach a consensus given any initial opinions because (1) the directed

network of influence is strongly connected (everyone is influenced by everyone at least indirectly) and (2) each node puts a positive weight on its opinion (we know that these two conditions imply there is t such that T^t is a positive matrix.)

3. (36 pts.) Consider the simple threshold model with the following hierarchical tree network in three layers.



Each node can choose either A or B . Initially B is a default choice at every node. Each node chooses to adopt A if and only if at least q fraction of its neighbors adopt A . All the nodes follow a myopic best response dynamics assumed in class.

Answer the following questions.

- Consider the cluster (subtree) that consists of node B1 and all the nodes “below” B1. What is the density of this cluster?
- Suppose that $q = 0.3$. If you can pick one node as an initial adopter of A to maximize the number of nodes to adopt A eventually (including the initial adopter), which node would you pick?
- What is the largest $q \in (0, 1)$ for which you can find one initial adopter to induce every node to adopt A eventually? Pick one such node.
- Interpret this as each node playing a coordination game against all its neighbors. Assume that payoff b from (B, B) is 2 and payoff a from (A, A) is 3 (this corresponds to the threshold $q = 0.4$). Suppose that each node has another option AB that works as A against A and B against B . However, it costs $c > 0$ to choose AB

each time. What is the largest cost c such that you can still induce every node to adopt A eventually by picking one initial adopter of A ?

Answer.

(a) (8 pts.) $2/3$

($2/3$ of $B1$'s neighbors ($C2$ and $C2$) are within the cluster. Every other node in the cluster has 100% neighbors within the cluster.)

(b) (8 pts.) Any C node ($C1, C2, C3, C4$).

(Note that, if there is only one initial adopter, then each node cannot face more than two A actions when playing B (because the network is a tree). This means that A (top node) can face at most $1/2$ A initially, any B node can face at most $1/3$ A initially, any C node can face at most $1/4$ A initially, and any D node can face 100% A initially in their neighbors. Since $q = 0.3$, each A or B or D node can be “infected” by A actions. So, we can maximize the number of A adopters by choosing any C node as the initial adopter of A , which infects three D nodes below, two B nodes, and A node.)

(c) (10 pts.)

$q = 0.25$. With this q , any node can be an initial adopter of A .

(This follows from the above analysis. The bottleneck is C node.)

(d) (10 pts.) $c = 1$.

(If $c \leq 3$, then AB becomes optimal with only just one A (with B for all the other nodes) in the neighborhood. If $c \leq 1$, then AB becomes optimal with only one AB (with B for all the other nodes) in the neighborhood. So AB spreads everywhere with one initial adopter of A if and only if $c \leq 1$. Once AB spreads, it becomes optimal to play A instead of AB .)